

Factors Contributing to Antibiotic Overprescription in Pediatric Upper Respiratory Tract Infections: Review Article

Mutah university

Introduction :

Upper respiratory tract infections (URTIs) are among the most common illnesses in children and represent a major reason for outpatient visits and healthcare utilization worldwide. Most pediatric URTIs, including the common cold, viral pharyngitis, and influenza, are caused by viruses and therefore do not require antibiotic treatment. Despite this, antibiotics continue to be prescribed frequently and often unnecessarily. Antibiotic overprescription has become a serious global public health concern because it contributes to the emergence of antimicrobial resistance, increases healthcare costs, exposes children to avoidable adverse drug reactions, and may disrupt the gut microbiota, potentially affecting immune system development. Understanding the factors that drive inappropriate antibiotic use is essential for developing effective strategies to promote rational prescribing. This review aims to explore the key factors influencing unnecessary antibiotic prescribing in pediatric upper respiratory tract infections.

Physician-related factors:

Prescribing behaviour is significantly influenced by physicians' responses to patients expectations, as well as their previous prescribing habits .Furthermore, fear of complications and parental blame, the child may not return for follow-up or limited access to medical care center, all these factors contribute to this practice . Notably, internal medicine and ENT clinics more likely to prescribe antibiotics compared to those in pediatric or other specialties.

Parent-related factors:

Parental pressure plays a major role in prescribing antibiotics because they

believe that antibiotics are necessary to shorten the recovery time even when the cause is viral. Additionally, caregiver and parents' educational background, areas of residence, income level and number of children significantly influence decision-making .

Diagnostic uncertainty :

Differentiating viral from bacterial infections remains challenging in clinical practice. This uncertainty often leads to precautionary antibiotic prescribing, especially in settings with poor hygiene or a higher risk of bacterial infections.

Healthcare system factors:

System-related issues, including the lack of clear guidelines, limited follow-up opportunities, and over-the-counter availability of antibiotics, significantly contribute to overprescription. Also, differences between public and private healthcare systems also influence prescribing patterns.

Conclusion:

Antibiotic overprescription in pediatric URIs is a multifactorial issue involving physician behavior, parental expectations, and healthcare system limitations. Addressing this problem requires the implementation of effective antibiotic stewardship programs. Hospitals and clinics should establish clear guidelines, improve physician training, and increase public awareness.

Furthermore, restrict Over-the-Counter Sales by enforcing policies preventing the sale of antibiotics without a prescription. A multifaceted approach involving healthcare providers, caregivers, policymakers, and public health organizations is required to reduce unnecessary antibiotic use and combat antimicrobial resistance. Education and communication (Educating patients that antibiotics do not work on viruses (colds, flu) and that they can cause unnecessary side effects like rashes, allergies, or diarrhea) is essential .